

Prime-Resolved Residual Fibers at a Fixed Shift: Exact Grouping SVD, Common Möbius Gauge, and Collision-Free Transversals

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Abstract

The literal fixed-shift residual map of a prime-squarefree orbit packet first restricts the full pre-terminal array to the inherited terminal equality support, multiplies a surviving input (m, r, j, γ) by $\mu(r)\Lambda(mj + h_0)$, and then identifies inputs with the same complete output key $(\gamma, d(m)r, j)$. Positive pre-terminal energy does not imply either terminal activation or survival under this identification. We determine the exact linear and arithmetic structure of the internal grouping gate, keeping these two losses separate.

For a general weighted fiber-summing operator we compute every singular value, its kernel, the Moore–Penrose inverse, and the canonical visible projection. Each active output fiber has exactly one visible direction; all remaining directions are collision modes. For a distinguished input, the grouped energy is an exact energy-weighted average of explicit fiber alignment coefficients. Conditional on separate terminal activation and downstream comparison gates, the remaining internal grouping statement is a lower bound on the angle of the physical array from a known kernel, not a uniform lower bound on the full coefficient space.

The fixed-shift arithmetic adds a more specialized fact. Assume $h_0 \neq 0$, $D = o(J)$, uniqueness of the declared row labels, and sufficiently large X . If the terminal prime $p = (mj + h_0)/r$ is retained, then the refined key $(\gamma, d(m)r, j, p)$ is injective on one declared source system. Hence after terminal restriction, residual grouping factors exactly into a collision-free prime-resolved lift and a final coherent sum over distinct terminal primes. Determinant ladders give the coefficient-blind upper bound $\ll_{h_0} X^{o(1)}(1 + L/R)$ for the maximal fiber size on a cofactor block $r \asymp R$. In the balanced range $R \gg L$, whenever the block has nonzero terminal diagonal energy, a rank-by-prime partition contains a collision-free subpacket with at least an $X^{-o(1)}$ fraction of that energy. This selection incurs no fixed positive-power loss inside that block; it is not a lower bound for the unreduced coherent sum.

Finally, on the inherited coprime squarefree equality support only, the source and cofactor signs combine as $\mu(d)\mu(r) = \mu(dr)$, so Möbius contributes one common phase on a residual fiber. Any remaining cancellation belongs to the actual source, mask, profile, and terminal amplitudes. No lower bound for the full fixed-shift coherent-prime square, parity improvement, or prime-pair consequence is claimed.

Keywords: residual grouping; terminal-prime resolution; fiber singular values; common Möbius gauge; determinant ladders; parity obstruction.

Convention. The shift $h_0 \neq 0$, one left-hand source system, and one direct residue/orbit slice are fixed. All physical domain and output norms use the inherited counting-measure atomic normalization. Terminal weights are displayed in the operator and are never absorbed into a redefined input norm. A collision means equality of the complete output key (γ, s, j) , not

Thus distinct points of one residual collision fiber necessarily carry distinct terminal primes. In the literal fixed- h_0 representation, “residual grouping” and “coherent terminal-prime synthesis” are not two independent many-to-one operations. The former is precisely the last operation that forgets the prime coordinate. This statement concerns the physical h_0 -slice; it does not identify the different coherent p -sum appearing after Fourier transformation of the completed frozen intercept family in TPC-48 [5].

1.2 The grouping spectrum is completely explicit

Let $q : \mathcal{U} \rightarrow \mathcal{Y}$ be any finite fiber map and

$$(Tz)_y = \sum_{q(u)=y} c_u z_u$$

between Hilbert spaces with positive atomic weights a_u and b_y . Put

$$S_y = \sum_{q(u)=y} \frac{|c_u|^2}{a_u}, \quad \rho_y^2 = b_y S_y.$$

We prove that the nonzero singular values of T are exactly the ρ_y . Every active fiber contributes one visible unit vector

$$(v_y)_u = \frac{\overline{c_u}}{a_u \sqrt{S_y}} \quad (q(u) = y)$$

and a kernel of codimension one. This gives the exact pseudoinverse, minimum-norm representative of every output, and quotient norm. On a restricted subspace, a lower bound is equivalent to a positive principal angle from this explicit kernel.

For one distinguished vector z , define

$$\eta_y(z) = \frac{\left| \sum_{q(u)=y} c_u z_u \right|^2}{S_y \sum_{q(u)=y} a_u |z_u|^2}.$$

Then $0 \leq \eta_y \leq 1$, and the grouped norm is exactly

$$\|Tz\|_b^2 = \sum_y \rho_y^2 \eta_y(z) \sum_{q(u)=y} a_u |z_u|^2.$$

After terminal activation has been proved and any downstream interface gates required by the target correlation have been supplied, the remaining internal grouping question is therefore an arithmetic lower bound for a displayed alignment statistic. There is no hidden spectral gap to infer from multiplicity or positive diagonal energy alone.

1.3 Balanced fibers admit a zero-cost transversal cover

Fix $y = (\gamma, s, j)$, write $s = dr$, and hold $d \mid s$ fixed. The admissible source prime ℓ and terminal prime p lie on the exact determinant ladder

$$rp - dj\ell = h_0.$$

If $g = (r, dj) \mid h_0$, all integral solutions are

$$p = p_0 + \frac{dj}{g}t, \quad \ell = \ell_0 + \frac{r}{g}t.$$

The inherited squarefree support has $(d, r) = 1$, so $g = (r, j) \mid h_0$. On a source-prime interval of length $O(L)$, the number of candidates for this d is $O_{h_0}(1 + L/r)$. Summing over $d \mid s$ gives the coefficient-blind upper bound

$$|\mathcal{F}_y(R)| \ll_{h_0} \tau(s) \left(1 + \frac{L}{R}\right) \quad (r \asymp R),$$

already implicit in the determinant-ladder boundary of TPC-45. Since $\tau(s) = X^{o(1)}$ on the polynomial-size support, balanced cofactors $R \gg L$ give subpower multiplicity.

Order the terminal primes in each full fiber and put the k -th smallest prime from every fiber into the k -th rank class. These classes form a deterministic, coefficient-independent partition into at most the maximal fiber size many collision-free transversals. If the terminal diagonal energy of the block is nonzero, one class captures at least the reciprocal of that number. Hence a balanced block with nonzero terminal diagonal energy contains a collision-free subpacket with an $X^{-o(1)}$ fraction of that energy. The selector is post-terminal and need not be a smooth Mellin cell. More importantly, a large collision-free summand does not lower-bound the sum of all rank classes: the omitted classes can still cancel it.

1.4 The Möbius gauge is common, not random

On the inherited coprime squarefree equality support, $d, r, s = dr$ are squarefree, $(d, r) = 1$, and

$$\mu(d)\mu(r) = \mu(s).$$

The source coefficient contains $\mu(d)$, while the terminal multiplier contains $\mu(r)$. Their product is therefore constant on a fixed residual fiber. This coherence was first isolated during the large-prime cofactor descent [4]; here it is inserted into the complete TPC-53 factor dictionary. The opposite-row phase and opposite profile are also common on a full (γ, s, j) -fiber. After those factors are removed, only the actual source coefficient, source profile, mask support, and positive-or-signed terminal shape remain in the fiber sum.

This is useful structural progress, but not a parity claim in the classical sieve-theoretic sense [2, Ch. 16]. The inherited theory supplies no uniform sector bound for the remaining complex amplitudes. We prove sharp phase-cone and magnitude-dominance criteria, and we prove that without either type of information the diagonal moduli alone may still form an exact polygonal cancellation.

1.5 Scope and order of the next gate

We use the program's level labels in the following precise sense. Level L0 consists of abstract finite-dimensional operator identities and inequalities that do not assert their arithmetic hypotheses for the literal packet. Level L1 consists of exact structural statements proved on the declared inherited fixed- h_0 support, including the prime-resolved injection and determinant-ladder upper bound. Level L2 would require a quantitative estimate reconnected to the full physical fixed-shift correlation, with activation, normalization, boundary, and all downstream comparison gates controlled. Nothing in this paper is L2.

The balanced transversal theorem is L1 but selects a post-terminal subpacket; it does not prove that the full physical output is large. No step identifies the residual low-sign Gram with the native affine Gram, changes the physical shift, or estimates a twin-prime correlation.

The result does, however, alter the practical order of attack. The next paper need not search for an abstract lower bound for a many-to-one map. After terminal activation and the remaining downstream interface requirements are kept explicit, it should estimate the terminal-prime coherence on one literal fiber or one declared rank interaction, with the source, cofactor, fixed shift, and actual atomic normalization all retained.

2 The literal fixed-shift map and its prime-resolved lift

2.1 Rows, coefficients, and the terminal-admissible set

We use the exact left-packet convention of TPC-53 [7]. Fix $h_0 \neq 0$, put $H = \text{rad } |h_0|$, and write a source row and an opposite row as

$$\begin{aligned} m &= d_m \ell_m, & m_\gamma &= d_\gamma \ell_\gamma, \\ \ell_m, \ell_\gamma &\in [L, 2L], & d_m, d_\gamma &\in [D, 2D], \\ (d_m, H) &= 1, & (d_\gamma, H) &= 1. \end{aligned} \quad (1)$$

with prime ℓ 's and squarefree d 's. The inherited separation makes an integer m determine (d_m, ℓ_m) on the declared cell. Write

$$Q = LD, \quad K = DJ, \quad QJ \asymp X, \quad \kappa_0 = \frac{1}{400}, \quad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \quad (2)$$

Fix one direct row-residue group, one orbit class $a \pmod{H_{\text{orb}}}$, one opposite-row box $\mathcal{Q}$, and one compact orbit interval $I_0 \subseteq (0, \infty)$. Here $H_{\text{orb}} = O_{h_0}(1)$ denotes the common orbit period; it is not being identified with $|h_0|$.

The weights ω_D and ψ_L range over the fixed bounded smooth dyadic families inherited from TPC-53: after the displayed normalization, their supports lie in $[D, 2D]$ and $[1, 2]$, respectively, and all normalized derivatives are uniformly bounded. We use the exact alias $\zeta_m := \zeta_m^{(1)}$. Thus the source coefficient is

$$a_m = \mu(d_m)(\log \ell_m)\omega_D(d_m)\psi_L(\ell_m/L)\zeta_m, \quad |\zeta_m| \ll 1. \quad (3)$$

The opposite row has unary factor $\eta_\gamma \mathcal{R}(q_\gamma)$, where

$$q_\gamma := \left(\log \frac{\ell_\gamma}{L}, \log \frac{d_\gamma}{D} \right), \quad |\eta_\gamma| = 1, \quad (4)$$

and $\mathcal{R}$ is the fixed smooth function denoted by R in TPC-53, relabeled here to avoid collision with the cofactor scale R . It is supported on the normalized opposite-row box. The direct orbit cell contributes a constant $\xi_a \neq 0$. The literal binary mask is

$$K_{m\gamma} = \mathbf{1}_{\ell_m \neq \ell_\gamma} \mathbf{1}_{|m - m_\gamma| > \Delta} \mathbf{1}_{(d_m, d_\gamma) \leq G}. \quad (5)$$

For the declared four-factor profile, set

$$\mathcal{W}_{m,\gamma}(j/J) = W_1(mj/X)\Psi_1(d_m j/K)W_2(m_\gamma j/X)\Psi_2(d_\gamma j/K), \quad (6)$$

where $W_1, W_2, \Psi_1, \Psi_2 \in C_c^\infty((0, \infty))$ are the fixed smooth cutoff functions of the declared packet.

Let $\mathcal{I}_X$ be the retained set of pairs (m, r) , let $\mathcal{O}_X$ be the complete opposite-row cell, and put

$$\mathcal{J}_{X,a} := \{j : j/J \in I_0, j \equiv a \pmod{H_{\text{orb}}}\}. \quad (7)$$

The full pre-terminal atomic universe is

$$\mathcal{U}^{\text{pre}} := \{(m, r, j, \gamma) : (m, r) \in \mathcal{I}_X, j \in \mathcal{J}_{X,a}, \gamma \in \mathcal{O}_X\}. \quad (8)$$

Thus the scalar coordinate of the lifted physical vector is

$$g_u := [G_X^L(m, r, j)]_\gamma = a_m K_{m\gamma} \eta_\gamma \mathcal{R}(q_\gamma) \xi_a \mathcal{W}_{m,\gamma}(j/J), \quad u = (m, r, j, \gamma). \quad (9)$$

The value is independent of r except for repetition over the retained cofactor support.

Let Φ_0 be the inherited terminal cutoff and define

$$\Lambda(n) := (\log n)\Phi_0(n/X). \quad (10)$$

This notation is a shaped prime amplitude, not the von Mangoldt function. For $u = (m, r, j, \gamma) \in \mathcal{U}^{\text{pre}}$, put

$$p_u := \frac{mj + h_0}{r} \quad (11)$$

whenever the quotient is integral. The terminal-admissible universe is the inherited coprime squarefree equality support

$$\begin{aligned} \mathcal{U}_{h_0}^{\text{term}} &:= \{u \in \mathcal{U}^{\text{pre}} : p_u \text{ is a prime in the inherited terminal support,} \\ &\quad p_u r - mj = h_0, \quad \Lambda(p_u r) \neq 0, \quad d_m, r, d_m r \text{ are squarefree,} \quad (d_m, r) = 1\}. \end{aligned} \quad (12)$$

Define separately the actually nonzero physical support

$$\mathcal{U}_{h_0}^{\text{phys}} := \{u \in \mathcal{U}_{h_0}^{\text{term}} : g_u \neq 0\}. \quad (13)$$

This distinction prevents pre-terminal rows from being silently treated as terminal-admissible atoms.

Thus, on every terminal-admissible atom, the inherited TPC-45/48 support conditions [4, 5] give

$$d_m, r, s = d_m r \text{ squarefree,} \quad (d_m, r) = 1, \quad \mu(d_m)\mu(r) = \mu(s). \quad (14)$$

The complete physical output key and its prime refinement are

$$q(u) = (\gamma, d_m r, j), \quad \tilde{q}(u) = (\gamma, d_m r, j, p_u). \quad (15)$$

2.2 Three exact operators

Use counting measure on every atomic coordinate. For an arbitrary array $g = (g_u)_{u \in \mathcal{U}^{\text{pre}}}$, the terminal diagonal is an operator on the full pre-terminal universe, defined by

$$(D_{h_0}g)_u := \begin{cases} \mu(r)\Lambda(p_u r)g_u, & u = (m, r, j, \gamma) \in \mathcal{U}_{h_0}^{\text{term}}, \\ 0, & u \in \mathcal{U}^{\text{pre}} \setminus \mathcal{U}_{h_0}^{\text{term}}. \end{cases} \quad (16)$$

Let $\tilde{\mathcal{Y}}_{h_0} = \tilde{q}(\mathcal{U}_{h_0}^{\text{term}})$. The prime-resolved lift is

$$(\mathcal{R}_{h_0}g)_{\tilde{y}} := \sum_{\substack{u \in \mathcal{U}_{h_0}^{\text{term}} \\ \tilde{q}(u) = \tilde{y}}} (D_{h_0}g)_u, \quad \tilde{y} \in \tilde{\mathcal{Y}}_{h_0}. \quad (17)$$

Finally, let Σ_p forget the prime coordinate:

$$(\Sigma_p x)_{\gamma, s, j} := \sum_{p: (\gamma, s, j, p) \in \tilde{\mathcal{Y}}_{h_0}} x_{\gamma, s, j, p}. \quad (18)$$

The exact terminal map of TPC-53 is

$$(\mathcal{T}_{h_0}g)_{\gamma, s, j} = \sum_{\substack{u = (m, r, j, \gamma) \in \mathcal{U}_{h_0}^{\text{term}} \\ s = d_m r}} \mu(r)\Lambda(p_u r)g_{m, r, j, \gamma}. \quad (19)$$

$$\mathcal{T}_{h_0} = \Sigma_p \mathcal{R}_{h_0}. \quad (20)$$

Every sum is restricted to the declared finite pre-terminal, terminal, and output supports. No projective layer or frozen intercept is present in these definitions.

2.3 Prime-resolved injectivity

The factorization becomes useful because $\mathcal{R}_{h_0}$ has no collision after restriction to the terminal-admissible coordinate space.

Lemma 2.1 (Fixed-coordinate fused-label injectivity). *Assume $D = o(J)$, $h_0 \neq 0$, and the row uniqueness in (1). For all sufficiently large X , if two rows in the same source system satisfy*

$$d_1(d_1\ell_1j + h_0) = d_2(d_2\ell_2j + h_0) \quad (21)$$

at the same retained $j \asymp J$, then $(d_1, \ell_1) = (d_2, \ell_2)$.

Proof. Rearranging gives

$$j(\ell_1d_1^2 - \ell_2d_2^2) = h_0(d_2 - d_1). \quad (22)$$

If the integer in parentheses on the left is nonzero, the left side has absolute value at least a fixed multiple of J , while the right side is $O_{h_0}(D) = o(J)$, a contradiction. Hence both sides vanish. Since $h_0 \neq 0$, first $d_1 = d_2$, and then $\ell_1 = \ell_2$. $\square$

Theorem 2.2 (Prime-resolved desingularization). *Assume $h_0 \neq 0$, $D = o(J)$, uniqueness of the row decomposition in (1), and sufficiently large X . On one declared source system, the map $\tilde{q} : \mathcal{U}_{h_0}^{\text{term}} \rightarrow \tilde{\mathcal{Y}}_{h_0}$ is injective. Consequently the restriction $\mathcal{R}_{h_0}|_{\ell^2(\mathcal{U}_{h_0}^{\text{term}})}$ is a weighted coordinate embedding. After the separate restriction from $\mathcal{U}^{\text{pre}}$ to $\mathcal{U}_{h_0}^{\text{term}}$, all many-to-one behavior of $\mathcal{T}_{h_0}$ occurs in the forget-prime map Σ_p .*

Proof. Suppose $\tilde{q}(u_1) = \tilde{q}(u_2)$. Then the two atoms have the same s, j, p, γ . Their fused labels satisfy

$$d_i(d_i\ell_ij + h_0) = d_i(pr_i) = p(d_ir_i) = ps.$$

Thus (21) holds, and lemma 2.1 gives the same source row. Then $r_i = s/d_i$ is also the same, so $u_1 = u_2$. Formula (20) shows that only Σ_p can identify distinct inputs. $\square$

Corollary 2.3 (Terminal primes parameterize a full collision fiber). *For $y = (\gamma, s, j)$, put*

$$\begin{aligned} \mathcal{F}_y^{\text{term}} &:= q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{term}}, & \mathcal{F}_y^{\text{phys}} &:= q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{phys}}, \\ \mathcal{P}_y^{\text{term}} &:= \{p_u : u \in \mathcal{F}_y^{\text{term}}\}, & \mathcal{P}_y^{\text{phys}} &:= \{p_u : u \in \mathcal{F}_y^{\text{phys}}\}. \end{aligned} \quad (23)$$

For compatibility below, write $\mathcal{F}_y := \mathcal{F}_y^{\text{term}}$ and $\mathcal{P}_y := \mathcal{P}_y^{\text{term}}$. Then $u \mapsto p_u$ is a bijection from $\mathcal{F}_y^{\text{term}}$ to $\mathcal{P}_y^{\text{term}}$, and its restriction is a bijection from $\mathcal{F}_y^{\text{phys}}$ to $\mathcal{P}_y^{\text{phys}}$. In particular, two distinct terminal-admissible atoms in one full output fiber carry distinct terminal primes.

2.4 The three energies

The preceding factorization keeps three norms separate. For an arbitrary lifted array g , define

$$\mathcal{E}_{\text{pre}}(g) := \sum_{u \in \mathcal{U}^{\text{pre}}} |g_u|^2, \quad (24)$$

$$\begin{aligned} \mathcal{D}_{h_0}(g) &:= \|D_{h_0}g\|_2^2 = \sum_{u \in \mathcal{U}_{h_0}^{\text{term}}} |\mu(r)|^2 |\Lambda(p_ur)|^2 |g_u|^2 \\ &= \sum_{u \in \mathcal{U}_{h_0}^{\text{term}}} |\Lambda(p_ur)|^2 |g_u|^2, \end{aligned} \quad (25)$$

$$\mathcal{G}_{h_0}(g) := \|\mathcal{T}_{h_0}g\|_2^2 = \sum_y \left| \sum_{u \in \mathcal{F}_y^{\text{term}}} \mu(r_u) \Lambda(p_ur_u) g_u \right|^2. \quad (26)$$

The last equality in $\mathcal{D}_{h_0}$ uses the inherited squarefree condition r on $\mathcal{U}_{h_0}^{\text{term}}$, so $|\mu(r)|^2 = 1$; it would be false on an unrestricted cofactor universe. The prime-resolved injection gives the exact identity

$$\|\mathcal{R}_{h_0}g\|_2^2 = \mathcal{D}_{h_0}(g). \quad (27)$$

It does not give $\mathcal{D}_{h_0} \asymp \mathcal{E}_{\text{pre}}$: that comparison also requires terminal activation and two-sided control of Λ on the selected support. Nor does it give a lower bound for $\mathcal{G}_{h_0}/\mathcal{D}_{h_0}$, because Σ_p creates the cross terms.

For the physical vector G_X^L , the pre-terminal energy is the raw TPC-53 lifted profile sum. It should not be confused with the normalized $1/(E_X J)$ model statistic used at an earlier intermediate stage in TPC-52 [6]. Equations (24)–(26) use literal counting-measure atoms throughout.

3 Exact singular-value geometry of a fiber sum

The terminal arithmetic will be inserted only after the finite-dimensional operator has been solved exactly.

3.1 Weighted atomic spaces

Let $\mathcal{U}$ and $\mathcal{Y}$ be finite sets, let $q : \mathcal{U} \rightarrow \mathcal{Y}$, and assign positive atomic weights $a_u > 0$ and $b_y > 0$. Write

$$\|z\|_a^2 = \sum_{u \in \mathcal{U}} a_u |z_u|^2, \quad \|w\|_b^2 = \sum_{y \in \mathcal{Y}} b_y |w_y|^2. \quad (28)$$

For coefficients $c_u \in \mathbb{C}$, define

$$(Tz)_y = \sum_{u \in F_y} c_u z_u, \quad F_y = q^{-1}(y). \quad (29)$$

Put

$$S_y := \sum_{u \in F_y} \frac{|c_u|^2}{a_u}, \quad \rho_y^2 := b_y S_y, \quad \mathcal{Y}_+ := \{y : S_y > 0\}. \quad (30)$$

Theorem 3.1 (Complete weighted fiber SVD). *The multiset of nonzero singular values of T is exactly*

$$\{\{\rho_y : y \in \mathcal{Y}_+\}\}, \quad (31)$$

with one occurrence contributed by each active output fiber. Thus distinct fibers with the same value of ρ_y contribute the corresponding singular-value multiplicity. In particular,

$$\text{rank } T = |\mathcal{Y}_+|, \quad \dim \ker T = |\mathcal{U}| - |\mathcal{Y}_+|. \quad (32)$$

For $y \in \mathcal{Y}_+$, the corresponding unit right and left singular vectors are

$$(v_y)_u = \begin{cases} \frac{\overline{c_u}}{a_u \sqrt{S_y}}, & u \in F_y, \\ 0, & u \notin F_y, \end{cases} \quad f_y = b_y^{-1/2} e_y, \quad (33)$$

and $Tv_y = \rho_y f_y$.

If P_y is the a -orthogonal projection onto $\mathbb{C}v_y$, then

$$(P_y z)_u = \frac{\overline{c_u}}{a_u S_y} \sum_{v \in F_y} c_v z_v \quad (u \in F_y), \quad (34)$$

and, with $P = \sum_{y \in \mathcal{Y}_+} P_y$,

$$\mathcal{H}_a(\mathcal{U}) = \ker T \oplus^{\perp_a} \text{Ran } P, \quad T^*T = \sum_{y \in \mathcal{Y}_+} \rho_y^2 P_y. \quad (35)$$

Proof. Conjugate T by the isometries $x_u = \sqrt{a_u}z_u$ and $W_y = \sqrt{b_y}w_y$. The resulting matrix has row

$$d_{y,u} = \sqrt{\frac{b_y}{a_u}} c_u \mathbf{1}_{q(u)=y}.$$

Distinct rows have disjoint supports, hence are orthogonal, and the squared norm of row y is $b_y S_y = \rho_y^2$. Normalizing the nonzero rows gives (33). The projection formula follows by taking the a -inner product with v_y . The remaining identities are the spectral decomposition and rank-nullity theorem. $\square$

Thus a collision fiber has no complicated internal spectrum: it has one visible direction and a completely flat zero spectrum in every transverse direction.

3.2 Kernel basis and coordinate transversals

Proposition 3.2 (Explicit collision kernel). *Fix $y \in \mathcal{Y}_+$ and choose $u_0 \in F_y$ with $c_{u_0} \neq 0$. Then*

$$k_{y,u} := c_u e_{u_0} - c_{u_0} e_u, \quad u \in F_y \setminus \{u_0\}, \quad (36)$$

is a basis of the kernel of the fiber functional $z \mapsto \sum_{u \in F_y} c_u z_u$. Hence a coordinate subspace supported on $S \subseteq \mathcal{U}$ is a domain of injectivity for T if and only if

$$c_u \neq 0 \quad (u \in S), \quad q|_S \text{ is injective.} \quad (37)$$

When these conditions hold, the singular values of the restriction are exactly

$$\left\{ |c_u| \sqrt{\frac{b_{q(u)}}{a_u}} : u \in S \right\}. \quad (38)$$

Proof. Every vector in (36) is killed because its two contributions are $c_{u_0} c_u - c_u c_{u_0} = 0$. The vectors are independent and their number is $|F_y| - 1$, which is the local nullity. If a coordinate restriction contains a zero coefficient or two directions from one active fiber, it has a kernel. Conversely, under (37) the restricted matrix has one nonzero entry in each occupied row and column, giving (38). $\square$

This necessary-and-sufficient statement is why a collision-free coordinate selection must retain at most one terminal atom from each complete output fiber. Merely bounding the fiber size does not make the unrestricted map injective.

3.3 Canonical quotient and the minimum-norm lift

We use the standard Moore–Penrose inverse convention [1].

Theorem 3.3 (Exact pseudoinverse). *For every $w \in \mathcal{H}_b(\mathcal{Y})$, the Moore–Penrose inverse is*

$$(T^\dagger w)_u = \begin{cases} \frac{\overline{c_u}}{a_u S_{q(u)}} w_{q(u)}, & q(u) \in \mathcal{Y}_+, \\ 0, & q(u) \notin \mathcal{Y}_+. \end{cases} \quad (39)$$

Let Q_+ denote the b -orthogonal projection onto the output coordinates indexed by $\mathcal{Y}_+$. Then

$$TT^\dagger = Q_+, \quad T^\dagger T = P, \quad (40)$$

and, for every w supported on $\mathcal{Y}_+$, $T^\dagger w$ is the unique minimum- a -norm solution of $Tz = w$, with

$$\inf_{Tz=w} \|z\|_a^2 = \sum_{y \in \mathcal{Y}_+} \frac{|w_y|^2}{S_y}. \quad (41)$$

If $\mathcal{Y}_+ \neq \emptyset$, then

$$\|T^\dagger\|_{b \rightarrow a} = \left(\min_{y \in \mathcal{Y}_+} \rho_y \right)^{-1}. \quad (42)$$

If $\mathcal{Y}_+ = \emptyset$, then $T = T^\dagger = 0$ and $\|T^\dagger\|_{b \rightarrow a} = 0$.

Proof. The formula is the inverse on each one-dimensional nonzero singular direction and zero on both the domain kernel and the inactive output coordinates. Direct substitution gives (40). Its input energy on one output coordinate is

$$\sum_{u \in F_y} a_u \left| \frac{\overline{c_u}}{a_u S_y} w_y \right|^2 = \frac{|w_y|^2}{S_y}.$$

Orthogonality of the fibers proves the energy and operator-norm formulas. When there is no active fiber, both operators are zero, which gives the last assertion. $\square$

Thus the quotient by the collision kernel has a canonical realization. Choosing one representative coordinate in each fiber is useful for support selection, but it is not the minimum-norm quotient unless the fiber already has one active coordinate.

3.4 Restricted subspaces and the exact missing angle

Assume throughout this subsection that $\mathcal{Y}_+ \neq \emptyset$. If $\mathcal{Y}_+ = \emptyset$, then $T = 0$, so no nonzero subspace can be a domain of injectivity for T .

Let

$$\rho_{\min} = \min_{y \in \mathcal{Y}_+} \rho_y, \quad \rho_{\max} = \max_{y \in \mathcal{Y}_+} \rho_y. \quad (43)$$

For a nonzero subspace $V \subseteq \mathcal{H}_a(\mathcal{U})$, define

$$\alpha(V) := \inf_{\substack{z \in V \\ \|z\|_a = 1}} \|Pz\|_a, \quad \gamma_T(V) := \inf_{\substack{z \in V \\ \|z\|_a = 1}} \|Tz\|_b. \quad (44)$$

Theorem 3.4 (Sharp angle criterion). *For every nonzero subspace V , in finite dimension,*

$$\alpha(V) > 0 \iff V \cap \ker T = \{0\}, \quad (45)$$

and

$$\rho_{\min} \alpha(V) \leq \gamma_T(V) \leq \rho_{\max} \alpha(V). \quad (46)$$

After normalizing the output coordinate y by $1/\rho_y$, the resulting operator $\hat{T}$ satisfies the exact identity

$$\|\hat{T}z\|_b = \|Pz\|_a, \quad \gamma_{\hat{T}}(V) = \alpha(V). \quad (47)$$

Proof. By [theorem 3.1](#),

$$\rho_{\min}^2 \|Pz\|_a^2 \leq \|Tz\|_b^2 \leq \rho_{\max}^2 \|Pz\|_a^2.$$

Taking the infimum on the unit sphere of V proves (46). Compactness of that sphere gives (45). Dividing each nonzero singular value by itself gives (47). $\square$

Pointwise injectivity on a sequence of changing subspaces is therefore not enough for a uniform theorem. The subspaces must remain quantitatively separated from the explicit collision kernel.

3.5 Alignment of a distinguished vector

For $z \in \mathcal{H}_a(\mathcal{U})$, put

$$E_y(z) := \sum_{u \in F_y} a_u |z_u|^2, \quad A_y(z) := \sum_{u \in F_y} c_u z_u, \quad (48)$$

and, when $S_y E_y(z) > 0$, define

$$\eta_y(z) := \frac{|A_y(z)|^2}{S_y E_y(z)}; \quad (49)$$

set $\eta_y = 0$ in the zero case.

Theorem 3.5 (Exact visible-energy identity). *For every z and every active fiber $y \in \mathcal{Y}_+$,*

$$0 \leq \eta_y(z) \leq 1, \quad \|P_y z\|_a^2 = \eta_y(z) E_y(z), \quad (50)$$

and

$$\boxed{\|Tz\|_b^2 = \sum_{y \in \mathcal{Y}_+} \rho_y^2 \eta_y(z) E_y(z)}. \quad (51)$$

On an active fiber, equality $\eta_y = 1$ holds precisely when the restriction of z to that fiber is a nonzero scalar multiple of v_y . Equality $\eta_y = 0$ holds precisely when that restriction lies in the fiber kernel, including the zero vector. On an inactive fiber $y \notin \mathcal{Y}_+$, one has $\eta_y = 0$ by convention, all coefficients c_u on F_y vanish, and the entire fiber is killed by T .

Proof. Cauchy–Schwarz in the weighted fiber gives $|A_y|^2 \leq S_y E_y$. Formula (34) gives $\|P_y z\|_a^2 = |A_y|^2 / S_y$. Multiplying by $\rho_y^2 = b_y S_y$ and summing gives (51). The equality cases are the equality case of Cauchy–Schwarz and the definition of the kernel. $\square$

For the physical terminal map, the inherited atomic norms have $a_u = b_y = 1$ and

$$c_u = \mu(r_u) \Lambda(p_u r_u). \quad (52)$$

The theorem must be applied with these literal coefficients. Choosing artificial weights $a_u = |c_u|^2$ would hide the terminal amplitude in a new domain norm and is not a physical normalization theorem.

4 Arithmetic anatomy of a full residual fiber

4.1 Divisor–prime parameterization

Fix an output $y = (\gamma, s, j)$ and write a source row as $m = d\ell$. Equations

$$s = dr, \quad pr - d\ell j = h_0 \quad (53)$$

give

$$r = \frac{s}{d}, \quad p = \frac{d^2 \ell j + h_0 d}{s}, \quad \ell = \frac{ps - h_0 d}{d^2 j}. \quad (54)$$

Since s is squarefree on the nonzero equality support, $d \mid s$ there automatically satisfies $(d, s/d) = 1$. To avoid replacing the declared physical carrier by a larger Diophantine model, define the exact admissible triple set

$$\mathfrak{A}_y := \left\{ (d, \ell, p) : \begin{array}{l} u = (d\ell, s/d, j, \gamma) \in \mathcal{U}_{h_0}^{\text{term}}, \quad q(u) = y, \\ d = d_{d\ell}, \quad \ell = \ell_{d\ell}, \quad p = p_u \end{array} \right\}, \quad (55)$$

and let

$$\mathfrak{D}_y^{\text{div}} := \{d : \text{there exist } \ell, p \text{ with } (d, \ell, p) \in \mathfrak{A}_y\}. \quad (56)$$

Writing $\mathcal{F}_y^{\text{term}} := \mathcal{F}_y$, the map

$$u = (m, r, j, \gamma) \mapsto (d_m, \ell_m, p_u) \quad (57)$$

is an exact bijection

$$\boxed{\mathcal{F}_y^{\text{term}} \longleftrightarrow \mathfrak{A}_y.} \quad (58)$$

Indeed, the reverse map sends (d, ℓ, p) to $(d\ell, s/d, j, \gamma)$, and membership in $\mathfrak{A}_y$ retains all source, residue, terminal, and carrier restrictions by definition. The physical fiber

$$\mathcal{F}_y^{\text{phys}} := \mathcal{F}_y \cap \mathcal{U}_{h_0}^{\text{phys}} \quad (59)$$

corresponds only to the subset of $\mathfrak{A}_y$ for which $g_u \neq 0$. Thus the terminal fiber must not be identified with its physical subset.

The projection of $\mathfrak{A}_y$ to p is one-to-one by [corollary 2.3](#). Projection to d need not be: one fixed divisor $d \in \mathfrak{D}_y^{\text{div}}$ can support several pairs of source and terminal primes. Thus $\tau(s)$ alone is not a fiber-size bound.

4.2 A fixed-divisor determinant ladder

Fix $d \mid s$, put $r = s/d$, and let

$$g = (r, dj). \quad (60)$$

Because $(d, r) = 1$, one has $g = (r, j)$. The elementary solution theory of the determinant equation gives the exact ladder.

Lemma 4.1 (Complete integral ladder). *The equation*

$$rp - dj\ell = h_0 \quad (61)$$

has an integral solution if and only if $g \mid h_0$. When one solution (p_0, ℓ_0) exists, all integral solutions are

$$\boxed{p = p_0 + \frac{dj}{g}t, \quad \ell = \ell_0 + \frac{r}{g}t, \quad t \in \mathbb{Z}.} \quad (62)$$

The two affine forms have nonzero cross determinant

$$\frac{dj}{g}\ell_0 - \frac{r}{g}p_0 = -\frac{h_0}{g}. \quad (63)$$

Proof. The gcd criterion is the standard solvability condition for a linear Diophantine equation. Differences of two solutions satisfy $r\Delta p = dj\Delta\ell$. Dividing by g gives $\Delta p = (dj/g)t$ and $\Delta\ell = (r/g)t$. Substitution of (p_0, ℓ_0) in the original equation gives the cross determinant. $\square$

Primality of both affine forms, fixed residue classes, the mask, and the smooth terminal support only delete ladder points. They do not make the ambient spacing shorter.

4.3 Exact ambient ladders and a coefficient-blind capacity upper bound

Let the source-prime support have diameter at most $C_L L$, with C_L fixed, and define the exact cofactor-block fiber

$$\mathcal{F}_y(R) := \{u \in \mathcal{F}_y : R \leq r_u \leq 2R\}. \quad (64)$$

Proposition 4.2 (Ambient-ladder capacity upper bound). *For every fixed (γ, s, j) , the number of terminal-admissible atoms with $R \leq r = s/d \leq 2R$ satisfies*

$$|\mathcal{F}_y(R)| \leq \sum_{\substack{d \in \mathfrak{D}_y^{\text{div}} \\ R \leq s/d \leq 2R \\ (s/d, j) | h_0}} \left(1 + \left\lfloor \frac{C_L L(s/d, j)}{s/d} \right\rfloor \right) \quad (65)$$

$$\ll_{h_0, C_L} \tau(s) \left(1 + \frac{L}{R} \right). \quad (66)$$

The same bound holds after imposing every literal prime, residue, mask, and profile support restriction.

Proof. For a fixed d , the ℓ -coordinate in (62) has spacing r/g . Hence an interval of diameter $C_L L$ contains at most $1 + \lfloor C_L L g / r \rfloor$ integral ladder points. Solvability forces $g = (r, j) \mid h_0$, so $g \leq |h_0|$. Sum over the eligible divisors and use $r \geq R$. Further arithmetic and support conditions only remove points. $\square$

The refined first line is more informative than the final divisor bound: it records which admissible divisors actually meet the cofactor block and the orbit gcd. Both lines are upper bounds. No matching lower bound, or even nonemptiness of a particular fiber, is asserted.

Corollary 4.3 (Balanced subpower multiplicity). *Suppose every residual label has polynomial size in X , and $R \geq cL$ for a fixed $c > 0$. Then*

$$M_X(R) := \max_y |\mathcal{F}_y(R)| = X^{o(1)}. \quad (67)$$

If $R = X^{\theta_R + o(1)}$ and $L = X^{\theta_L + o(1)}$, the coefficient-blind polynomial capacity cost is at most

$$\mu_{\text{cap}} = (\theta_L - \theta_R)_+. \quad (68)$$

Proof. The standard divisor bound [3, Ch. 1] gives $\tau(s) = X^{o(1)}$ uniformly for polynomial-size s . Apply proposition 4.2. At the exponent level, $1 + L/R = X^{(\theta_L - \theta_R)_+ + o(1)}$. $\square$

An upper-bound sieve may remove logarithmic powers by requiring the two ladder forms to be prime. Such a count does not by itself remove the positive-power term L/R from this ambient coefficient-blind upper bound, furnish a matching lower bound, or estimate the actual complex amplitudes on the surviving points. Those are different tasks.

4.4 A local separation certificate

The prime parameterization also gives a simple collision-separation identity that does not use h_0 .

Proposition 4.4 (Scaled-prime gap). *If two distinct inputs $u_1, u_2 \in \mathcal{F}_{\gamma, s, j}$ have rows $m_i = d_i \ell_i$ and terminal primes p_i , then*

$$\frac{p_1}{d_1} - \frac{p_2}{d_2} = \frac{j(m_1 - m_2)}{s}, \quad \left| \frac{p_1}{d_1} - \frac{p_2}{d_2} \right| \geq \frac{j}{s}. \quad (69)$$

Consequently, inside this fixed output fiber, any interval in the coordinate p/d of diameter strictly less than j/s contains at most one input atom.

Proof. Since $r_i = s/d_i$, the terminal equation gives

$$\frac{p_i}{d_i} = \frac{m_i j + h_0}{s}.$$

Subtract. Distinct inputs have distinct rows by theorem 2.2, so $|m_1 - m_2| \geq 1$. $\square$

This is a deterministic local certificate, not an economical global cover: the number of p/d -intervals needed over the entire physical range can have a positive polynomial size. The rank construction in the next section uses the actual fiber multiplicity instead.

5 Canonical collision-free transversals

The exact multiplicity bound can be converted into a selection theorem without assigning free phases or changing the physical atomic weights.

5.1 Rank coloring of the terminal primes

Assume $\mathcal{U}_{h_0}^{\text{term}} \neq \emptyset$. For each nonempty terminal output $y = (\gamma, s, j)$, order its terminal-prime set from [corollary 2.3](#):

$$p_1(y) < p_2(y) < \cdots < p_{n_y}(y), \quad n_y = |\mathcal{F}_y|. \quad (70)$$

Let $u_k(y)$ be the unique input with terminal prime $p_k(y)$, and put

$$M = \max_y n_y, \quad \mathcal{S}_k := \{u_k(y) : n_y \geq k\}, \quad 1 \leq k \leq M. \quad (71)$$

The construction depends only on the physical terminal prime and a fixed ordering rule, not on the coefficient values.

Theorem 5.1 (Canonical prime-rank cover). *The terminal-admissible input set is the disjoint union*

$$\mathcal{U}_{h_0}^{\text{term}} = \bigsqcup_{k=1}^M \mathcal{S}_k, \quad (72)$$

and $q|_{\mathcal{S}_k}$ is injective for every k . Conversely, a partition of the input coordinates into collision-free coordinate sets requires at least M sets whenever a fiber of size M is present. Thus the number of rank classes is optimal.

Proof. Every input has one rank inside its fiber, proving the disjoint union. A fixed rank contributes at most one input to each fiber, so q is injective there. A fiber with M coordinates forces its members into different collision-free sets, proving optimality. $\square$

For any nonnegative atomic energy e_u , write

$$E_k = \sum_{u \in \mathcal{S}_k} e_u, \quad E = \sum_u e_u. \quad (73)$$

Corollary 5.2 (Sharp energy capture). *At least one rank class satisfies*

$$E_k \geq \frac{E}{M}. \quad (74)$$

The constant $1/M$ is optimal for arbitrary nonnegative atomic data.

Proof. Sum the M rank energies. Equality is possible when all $E_k = E/M$. $\square$

5.2 Specialization to terminal diagonal energy

Let

$$x = D_{h_0} g, \quad e_u = |x_u|^2 = |\Lambda(p_u r_u)|^2 |g_u|^2. \quad (75)$$

On a collision-free rank class, grouping creates no cross term:

$$\|\mathcal{T}_{h_0}(g \mathbf{1}_{\mathcal{S}_k})\|_2^2 = \sum_{u \in \mathcal{S}_k} |x_u|^2 = E_k. \quad (76)$$

For a cofactor scale R , define the exact terminal block

$$\mathcal{B}_R := \{u \in \mathcal{U}_{h_0}^{\text{term}} : R \leq r_u \leq 2R\}, \quad g_R := g \mathbf{1}_{\mathcal{B}_R}, \quad (77)$$

and, using (64), its maximal fiber size

$$\mathcal{F}_y(R) = \mathcal{F}_y \cap \mathcal{B}_R, \quad M_X(R) := \max_y |\mathcal{F}_y(R)|. \quad (78)$$

Within each nonempty $\mathcal{F}_y(R)$, order the terminal primes and let $\mathcal{S}_k(R)$ contain the atom of block rank k , whenever it exists. Thus $\{\mathcal{S}_k(R) : 1 \leq k \leq M_X(R)\}$ is the prime-rank cover applied to $\mathcal{B}_R$, rather than the restriction of a possibly different global rank ordering.

Theorem 5.3 (Balanced collision-free block subpacket). *Assume that every residual label s occurring in $\mathcal{B}_R$ has polynomial size in X , and let g be a physical array such that $\mathcal{D}_{h_0}(g_R) > 0$. There is a canonical family of*

$$M_X(R) \ll_{h_0, C_L} X^{o(1)} \left(1 + \frac{L}{R}\right) \quad (79)$$

collision-free prime-rank classes covering every terminal-admissible atom in that block. At least one block rank class satisfies

$$\|\mathcal{T}_{h_0}(g_R \mathbf{1}_{\mathcal{S}_k(R)})\|_2^2 \geq \frac{\mathcal{D}_{h_0}(g_R)}{M_X(R)}. \quad (80)$$

In particular, if $R \geq cL$, then conditional on nonzero block terminal diagonal energy, the selected collision-free block subpacket incurs no fixed positive-power loss relative to that block energy.

Proof. Combine proposition 4.2 and theorem 5.1. Apply corollary 5.2 to the physical terminal diagonal weights of g_R in (75), then use (76). $\square$

This is a positive post-terminal selection result on a specified nonzero block. It says that the balanced block terminal diagonal admits a coordinate subpacket with no fixed positive-power selection cost, on which physical grouping is exactly isometric after the literal terminal multiplier has been applied. It does not assert that the block energy is nonzero, compare that energy with pre-terminal or global energy, or say that a preassigned rank captures it: the family is canonical, while the choice of its energy-rich member may depend on g_R .

5.3 Why the selected lower bound does not return to the full sum

Write

$$X_k := \mathcal{T}_{h_0}(g \mathbf{1}_{\mathcal{S}_k}), \quad \mathcal{T}_{h_0}g = \sum_{k=1}^M X_k. \quad (81)$$

The X_k occupy the same output coordinate system and are not orthogonal. Therefore $\|X_k\|^2 \leq \|\sum_l X_l\|^2$ is false in general.

For one selected class $S = \mathcal{S}_k$, put

$$D_S = \sum_{u \in S} |x_u|^2, \quad D_R = \sum_{u \notin S} |x_u|^2. \quad (82)$$

Every remaining rank class is also collision-free. The triangle and Cauchy–Schwarz inequalities give the valid one-sided estimate

$$\|\mathcal{T}_{h_0}g\|_2 \geq \left(\sqrt{D_S} - \sqrt{(M-1)D_R} \right)_+. \quad (83)$$

Indeed, the sum of the norms of the other $M-1$ rank outputs is at most $\sqrt{M-1}\sqrt{D_R}$. Unless the selected rank dominates its complement at this scale or the inter-rank phases are controlled, (83) can be zero.

The obstruction is real already for a two-atom fiber: two equal terminal amplitudes with opposite remaining phases give positive diagonal energy in both rank classes and zero full output. Hence theorem 5.3 cannot be upgraded by monotonicity.

5.4 Selection versus the canonical quotient

The rank class is a sparse coordinate representative of each occupied fiber. The minimum-norm representative from [theorem 3.3](#) instead distributes an output over all fiber coordinates in proportion to $\bar{c}_u/a_u$. These solve different problems:

- (i) the pseudoinverse is canonical for reconstructing an already known grouped output;
- (ii) a rank transversal is canonical for eliminating collisions before a lower-bound argument; and
- (iii) neither construction supplies the phase relation among all rank outputs of the distinguished physical array.

The rank selector is also a post-terminal arithmetic selector. It need not be representable by a fixed smooth source/Mellin box, and the present paper does not move earlier profile estimates through it. Any later analytic use must verify stability of the relevant estimates on this exact subset or work directly with the full coherence statistic below.

6 The physical fiber: common gauges and the exact coherence gate

6.1 Extraction of every common factor

Fix a physical output $y = (\gamma, s, j)$ with nonzero terminal diagonal energy. For $u = (m, r, j, \gamma) \in \mathcal{F}_y^{\text{phys}}$, write $m = d\ell$. Combining [\(3\)](#), [\(9\)](#), and [\(14\)](#) gives

$$\mu(r)\mu(d) = \mu(s). \quad (84)$$

Moreover, γ, s, j, a are fixed on the fiber. Define the common output factor

$$\Omega_y := \mu(s)\eta_\gamma \mathcal{R}(q_\gamma)\xi_a W_2(m_\gamma j/X)\Psi_2(d_\gamma j/K), \quad (85)$$

and the remaining source-terminal amplitude

$$b_u := (\log \ell)\omega_D(d)\psi_L(\ell/L)\zeta_m K_{m_\gamma} W_1(mj/X)\Psi_1(dj/K)\Lambda(mj + h_0). \quad (86)$$

The equality $\Lambda(p_u r) = \Lambda(mj + h_0)$ has been used.

Proposition 6.1 (Literal fiber gauge factorization). *For every $u \in \mathcal{F}_y^{\text{phys}}$,*

$$(D_{h_0} G_X^L)_u = \Omega_y b_u. \quad (87)$$

Consequently

$$(\mathcal{T}_{h_0} G_X^L)_y = \Omega_y \sum_{u \in \mathcal{F}_y^{\text{phys}}} b_u, \quad (88)$$

$$\mathcal{E}_y^{\text{diag}} := \sum_{u \in \mathcal{F}_y^{\text{phys}}} |(D_{h_0} G_X^L)_u|^2 = |\Omega_y|^2 \sum_{u \in \mathcal{F}_y^{\text{phys}}} |b_u|^2. \quad (89)$$

In particular, the product of the two Möbius factors is the common sign $\mu(s)$ on one full residual fiber; the individual factors $\mu(d)$ and $\mu(r)$ may vary. The opposite-row unit phase is also common on the fiber.

Proof. Substitute the literal factors. The source sign $\mu(d)$ and terminal sign $\mu(r)$ combine by [\(84\)](#). All opposite-row factors depend only on (γ, j) , and the direct orbit multiplier depends only on the fixed class a . The remaining factors are exactly [\(86\)](#). Squaring before summation gives [\(89\)](#). $\square$

This extends the common-Möbius-phase observation of TPC-45 to the exact four-factor packet. It rules out one tempting but incorrect heuristic: independent Möbius randomness inside a fixed residual fiber cannot be the source of a grouping gain or loss. The amplitude b_u still contains the inherited coefficient ζ_m , source shapes, the literal mask, and terminal amplitude, and these are not proved to have a common phase.

6.2 Terminal coherence and the cross-term identity

For an arbitrary terminalized array $x = D_{h_0}g$, define on every nonzero terminal fiber

$$D_y(x) := \sum_{u \in \mathcal{F}_y^{\text{term}}} |x_u|^2, \quad \chi_y(x) := \frac{\left| \sum_{u \in \mathcal{F}_y^{\text{term}}} x_u \right|^2}{D_y(x)}. \quad (90)$$

Put $\chi_y = 0$ when $D_y = 0$.

Theorem 6.2 (Exact terminal-coherence identity). *If $n_y = |\{u \in \mathcal{F}_y^{\text{term}} : x_u \neq 0\}|$, then*

$$0 \leq \chi_y(x) \leq n_y, \quad (91)$$

and

$$\|\mathcal{T}_{h_0}g\|_2^2 = \sum_y \chi_y(x) D_y(x) \quad (92)$$

$$= \mathcal{D}_{h_0}(g) + 2 \operatorname{Re} \sum_y \sum_{\substack{u < v \\ u, v \in \mathcal{F}_y^{\text{term}}}} x_u \overline{x_v}. \quad (93)$$

The upper endpoint is sharp for every $n_y \geq 1$. The lower endpoint is sharp for $n_y \geq 2$; a nonzero one-atom fiber has $\chi_y = 1$.

Proof. The upper bound is Cauchy–Schwarz and the lower bound is positivity. Expanding each squared fiber sum gives (92) and (93). Equal nonzero phases and equal moduli give $\chi_y = n_y$. When $n_y \geq 2$, a nonzero vector in the fiber kernel gives $\chi_y = 0$; when $n_y = 1$, the quotient is identically one. $\square$

For $\mathcal{D}_{h_0}(g) > 0$, define the global physical grouping coherence

$$\chi_{\text{grp}}(g) := \frac{\|\mathcal{T}_{h_0}g\|_2^2}{\mathcal{D}_{h_0}(g)} = \frac{\sum_y \chi_y D_y}{\sum_y D_y}. \quad (94)$$

When $\mathcal{D}_{h_0}(g) = 0$, put $\chi_{\text{grp}}(g) := 0$. This convention is compatible with the zero terminal output. This number can be zero, one, or larger than one. It is the exact ratio needed when terminal diagonal energy, rather than pre-terminal energy, is the input normalization.

The alignment from [theorem 3.5](#) uses

$$E_y(g) := \sum_{u \in \mathcal{F}_y^{\text{term}}} |g_u|^2, \quad S_y := \sum_{u \in \mathcal{F}_y^{\text{term}}} |\Lambda(p_u r_u)|^2. \quad (95)$$

Whenever $S_y E_y(g) D_y(D_{h_0}g) > 0$, the two diagnostics are related by

$$\eta_y(g) = \chi_y(D_{h_0}g) \frac{D_y(D_{h_0}g)}{S_y E_y(g)}. \quad (96)$$

In the zero cases both diagnostics retain their separately declared conventions; the displayed quotient is not used. They should not be interchanged when terminal weights are nonuniform.

6.3 Two sharp positive criteria

The exact coherence can be bounded below if new phase or magnitude information is supplied.

Theorem 6.3 (Sector criterion). *Suppose that for every active fiber y there is an angle ϑ_y such that*

$$|\arg(e^{-i\vartheta_y} x_u)| \leq \theta < \frac{\pi}{2} \quad (u \in \mathcal{F}_y^{\text{term}}, x_u \neq 0). \quad (97)$$

Then

$$\left| \sum_{u \in \mathcal{F}_y^{\text{term}}} x_u \right|^2 \geq \cos^2 \theta \left(\sum_{u \in \mathcal{F}_y^{\text{term}}} |x_u| \right)^2 \geq \cos^2 \theta D_y(x), \quad (98)$$

and hence

$$\|\mathcal{T}_{h_0} g\|_2^2 \geq \cos^2 \theta \mathcal{D}_{h_0}(g). \quad (99)$$

For $\theta = 0$, all amplitudes lie on one ray and grouping cannot reduce terminal diagonal energy.

Proof. After rotation by $e^{-i\vartheta_y}$, the real part of each amplitude is at least $\cos \theta$ times its modulus. The modulus of the sum is at least its real part. Sum over the fibers. $\square$

The common gauge in [proposition 6.1](#) removes $\mu(s)\eta_\gamma$ before testing (97). It does not prove that the remaining b_u satisfy the sector condition.

Theorem 6.4 (Sharp magnitude-only lower envelope). *Fix $n \geq 1$ and nonnegative magnitudes $r_1, \dots, r_n$, put*

$$L_1 = \sum_{i=1}^n r_i, \quad R_1 = \max_i r_i. \quad (100)$$

Then

$$\min_{\theta_1, \dots, \theta_n \in \mathbb{R}} \left| \sum_{i=1}^n r_i e^{i\theta_i} \right| = (2R_1 - L_1)_+, \quad \max_{\theta_1, \dots, \theta_n \in \mathbb{R}} \left| \sum_{i=1}^n r_i e^{i\theta_i} \right| = L_1. \quad (101)$$

Thus magnitudes alone force a positive fiber lower bound if and only if one term strictly dominates the sum of all the others.

Proof. The triangle inequality gives the maximum and the lower bound $R_1 - (L_1 - R_1) = 2R_1 - L_1$. If $R_1 \geq L_1 - R_1$, place every smaller segment opposite the largest to attain that difference. If $R_1 \leq L_1 - R_1$, the polygon inequality permits the segments to close, attaining zero. $\square$

This theorem is a sharp non-implication for magnitude-based selection. Unless a dominance inequality is proved, the moduli selected in TPC-54 do not imply a lower bound uniformly over their possible phases. It does not assert that the fixed physical phases realize the minimizing polygon.

6.4 Exact gate factorization

For any selected pre-terminal packet with $\mathcal{E}_{\text{pre}}(g) > 0$, define

$$\delta_{\text{term}}(g) := \frac{\mathcal{D}_{h_0}(g)}{\mathcal{E}_{\text{pre}}(g)}. \quad (102)$$

Combining definitions gives the exact ratio

$$\boxed{\frac{\|\mathcal{T}_{h_0} g\|_2^2}{\mathcal{E}_{\text{pre}}(g)} = \delta_{\text{term}}(g) \chi_{\text{grp}}(g)}. \quad (103)$$

The two factors ask different questions:

- (i) δ_{term} measures which pre-terminal atoms meet a prime and a nonzero terminal cutoff, together with the literal size of Λ ;
- (ii) χ_{grp} measures coherent addition after those terminal atoms are resolved.

Neither is supplied by pre-terminal carrier coverage. The balanced transversal theorem obtains $\chi_{\text{grp}} = 1$ on one selected rank class relative to that class's terminal diagonal, at the cost of deleting the other ranks. For the undeleted physical vector, the second factor in (103) remains the next arithmetic gate.

7 Endpoint ledger, claim boundary, and the next gate

7.1 The capacity cost at the inherited endpoint

The declared row scales inherited from TPC-53 [7] are

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad D = X^{10049/52500+o(1)}. \quad (104)$$

Therefore

$$L = \frac{Q}{D} = X^{99979/210000+o(1)}. \quad (105)$$

On a cofactor block $R = X^{\theta_R+o(1)}$, the worst-case transversal-selection exponent certified by corollary 4.3 is

$$\mu_{\text{cap}} = \left(\frac{99979}{210000} - \theta_R \right)_+. \quad (106)$$

Thus every fixed balanced range $R \gg L$ has

$$M_X(R) = X^{o(1)}, \quad \mu_{\text{cap}} = 0. \quad (107)$$

This means that the rank cover has no fixed positive-power loss inside a balanced block. It is not a claim that terminal diagonal energy is concentrated on, or even nonzero in, such a block.

At the older cofactor scale $R \asymp J$, the coefficient-blind ladder certificate gives the worst-case factor

$$\frac{L}{R} = X^{15077/105000+o(1)}. \quad (108)$$

The exponent in this support-only certificate exceeds the nominal endpoint allowance $X^{1/400+o(1)}$ by

$$\frac{15077}{105000} - \frac{1}{400} = \frac{29629}{210000} > 0. \quad (109)$$

Thus that certificate alone cannot close the endpoint budget at $R \asymp J$. The actual physical multiplicity may be smaller; no matching lower bound is asserted. Primality can give logarithmic thinning but does not improve this support-only power count.

7.2 A complete loss ledger

Suppose a parent physical atomic energy is first restricted to a declared compatible packet with fraction

$$X^{-\lambda_{\text{sel}}+o(1)}, \quad (110)$$

and then to the pre-terminal cofactor block

$$\mathcal{B}_R^{\text{pre}} := \{u \in \mathcal{U}^{\text{pre}} : R \leq r_u \leq 2R\}, \quad g_R^{\text{pre}} := g \mathbf{1}_{\mathcal{B}_R^{\text{pre}}}, \quad (111)$$

with energy fraction

$$X^{-\lambda_{\text{blk}}+o(1)}. \quad (112)$$

Suppose first that terminal activation on this full pre-terminal block satisfies

$$\delta_{\text{term}}(g_R^{\text{pre}}) \geq X^{-\lambda_{\text{term}}+o(1)}. \quad (113)$$

The resulting terminal diagonal energy is then partitioned by the block prime-rank classes of [theorem 5.3](#). Selecting the energy-rich collision-free rank costs at most the certified exponent μ_{cap} . The selected-rank grouped energy therefore has the formal lower exponent

$$\lambda_{\text{sel}} + \lambda_{\text{blk}} + \mu_{\text{cap}} + \lambda_{\text{term}}. \quad (114)$$

This conclusion is relative to the selected post-terminal rank. The order is essential: terminal activation is established on the pre-terminal block before the rank pigeonhole is applied to its terminal diagonal energy. The selected-rank statement does not return to the full output.

For the undeleted physical terminal vector of the same block, instead suppose one proves

$$\chi_{\text{grp}} \geq X^{-\lambda_{\text{coh}}+o(1)}. \quad (115)$$

Then the exact factorization [\(103\)](#) gives the combined exponent

$$\lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{coh}}. \quad (116)$$

Any later native-Gram comparison, frozen-family localization, boundary truncation, or ultra-tail reconnection must add its own nonnegative loss. To remain within the nominal TPC endpoint budget, the complete eventual sum must satisfy

$$\boxed{\lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{coh}} + \lambda_{\text{loc}} + \lambda_{\text{Gram}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}} \leq \frac{1}{400}.} \quad (117)$$

At equality, a further error of size $X^{-1/400+o(1)}$ need not be absorbed; strict slack or a constant-level comparison is the safe closure condition.

The present paper proves none of λ_{blk} , λ_{term} , λ_{coh} , or the downstream losses for the full inherited packet. It proves $\mu_{\text{cap}} = 0$ for the coordinate-selection problem conditional on a declared balanced block with nonzero terminal diagonal energy.

7.3 Six-point physical audit

The interfaces used above satisfy the following audit.

- (A1) *Literal coefficients.* The source coefficient, binary mask, four smooth profile factors, cofactor sign, and terminal amplitude are displayed in [\(3\)](#), [\(5\)](#), [\(6\)](#), and [\(16\)](#).
- (A2) *Fixed physical shift.* Every arithmetic equation uses the one fixed h_0 . The frozen Fourier family is mentioned only to distinguish it from this map.
- (A3) *Atomic normalization.* Domain, prime-resolved output, and grouped output all use inherited counting measure. The terminal coefficient is not hidden in the norm.
- (A4) *Canonical representation.* The map is the literal TPC-53 coordinate map. No projective representation or favorable Mellin layer is selected.
- (A5) *Actual support.* The full pre-terminal universe, terminal-admissible subset, and physically nonzero subset are separated in [\(8\)](#), [\(12\)](#), and [\(13\)](#).
- (A6) *Endpoint budget.* The only unconditional selection exponent is the explicit μ_{cap} in [\(106\)](#); all unproved losses remain visible in [\(117\)](#).

7.4 Claim table

Statement	Status
Weighted fiber SVD, kernel, pseudoinverse, and angle criterion	Unconditional finite-dimensional theorem (L0).
Prime-resolved injection	Exact on one declared fixed- h_0 physical source system under the inherited $D = o(J)$ row geometry (L1).
Determinant-ladder capacity	Exact coefficient-blind upper bound on the declared squarefree terminal support (L1).
Balanced rank cover	Canonical post-terminal cover and existence of a collision-free subpacket with an $X^{-o(1)}$ fraction of nonzero block terminal diagonal energy; no block-occupancy estimate (L1).
Common Möbius-gauge factorization	Exact common product $\mu(d)\mu(r) = \mu(s)$ on the inherited coprime squarefree support; no claim about the remaining complex amplitudes (L1).
Terminal activation	Not proved for the full pre-terminal packet.
Full coherent output lower bound	Not proved; exactly equivalent to a lower estimate for the displayed physical coherence/alignment together with terminal activation.
Parity or twin-prime consequence	Not claimed.

In particular, the following implications remain forbidden:

- (i) subpower maximal multiplicity does not imply a lower bound for the unselected full sum;
- (ii) common Möbius phase does not imply a sector condition for ζ_m and the source/mask profile;
- (iii) a collision-free selected rank need not be a compatible smooth pre-terminal cell;
- (iv) refined-key, equivalently fixed (s, j, p, γ) , injectivity does not make fixed (s, j, γ) injective, because different terminal primes are exactly the collision modes;
- (v) the physical fixed- h_0 prime synthesis is not the frozen-intercept Fourier p -sum; and
- (vi) a residual output norm is not automatically the native affine physical Gram face.

7.5 The next gate

The next paper should keep one literal full fiber or one explicitly declared cofactor block and study

$$\sum_{u \in \mathcal{F}_{\gamma, s, j}^{\text{phys}}} b_u, \quad (118)$$

with b_u given by (86). There are three honest doors:

- Door A.** prove a deterministic sector, dominance, or restricted-Gram lower bound for the distinguished physical amplitudes on one nonempty literal regime;
- Door B.** prove a two-sided coherent terminal-prime square after an explicitly labeled averaging or smoothing, then quantify exactly why that model does or does not localize back to fixed h_0 ; or
- Door C.** construct a literal admissible packet whose remaining amplitudes approach the collision kernel, giving a coefficient-specific saturation or no-go theorem for this route.

An estimate for an arbitrary smooth proxy, a random-phase model, or an averaged shift would be useful only under its own label. It would not be reported as a fixed-shift parity gain.

Conclusion. The residual-grouping obstruction is now explicitly identified at the operator and support levels. Under the declared hypotheses, the refined terminal key is injective and the prime-resolved operator restricted to terminal-admissible coordinates is a weighted coordinate embedding; forgetting the terminal prime creates rank-one output fibers with an explicit collision kernel. Balanced fibers admit an optimal subpower transversal cover, and the product of the two literal Möbius signs is common inside each fiber. After terminal activation is controlled, the remaining internal grouping question is how the distinguished source–mask–prime amplitudes align with the visible direction. Fixed-shift localization, native-Gram comparison, boundary control, and tail reconnection remain separate downstream gates. Every one of these steps is strictly upstream of a parity-sensitive or twin-prime conclusion.

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